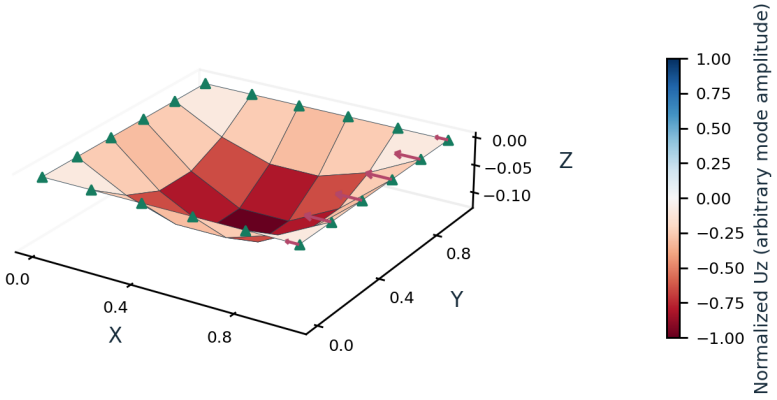
Simply supported plate buckling

TG_plate_uniaxial_buckling | SOL 105 | Independent analytical verification

Square plate a=b=1 m, t=0.01 m, E=210 GPa, nu=0.3; 5x5 shell mesh.

Geometry / mesh	36 nodes; 25 elements. Square plate a=b=1 m, t=0.01 m, E=210 GPa, nu=0.3; 5x5 shell mesh.
Material / section	E=2.1e+11, nu=0.3; shell thickness=0.01. SI: m, N, kg, s, Pa
Boundary conditions	All edges transverse displacement fixed; left edge axial displacement fixed; one Y anchor; uniform 1 Pa compressive axial stress.
Applied load	Uniform 1 Pa reference compression, applied as consistent edge nodal forces; multiplier equals critical stress numerically in Pa.

Mode 1 | load factor = 7.7186e+07



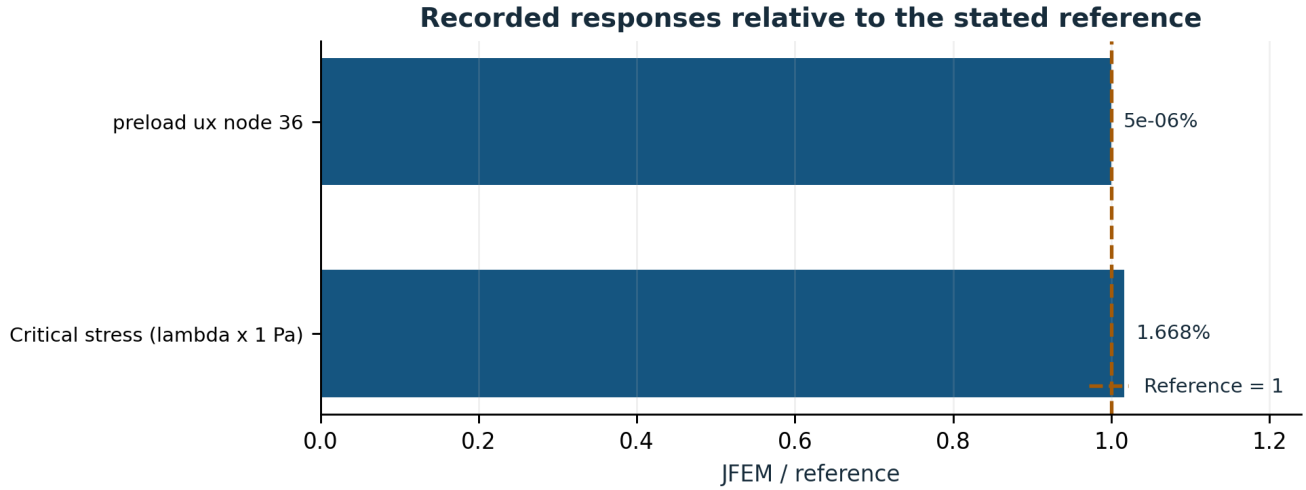
Shape amplified x0.396; grey = original. Eigenvector sign and amplitude are arbitrary.

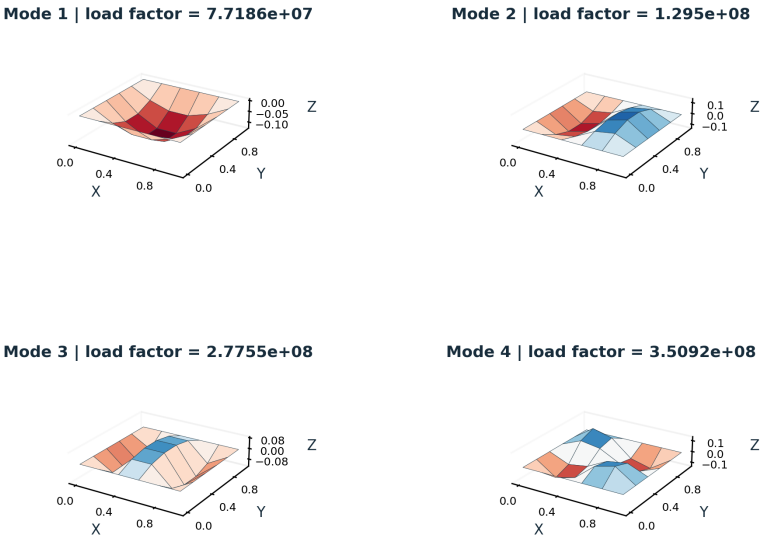
Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
Critical stress (lambda x 1 Pa) [GPa]	0.077186	0.07592	1.668%	PASS
preload ux node 36 [m]	-4.7619e-12	-4.7619e-12	5e-06%	REGRESSION

Evidence, convergence and interpretation





Independently normalized eigenvectors; sign and amplitude arbitrary. Grey = original.

First available modes. Red/blue indicate opposite signed components; each mode is normalized independently.

The supplied input omitted SPC selection from the buckling subcase. This rerun explicitly adds SPC=1 to that subcase, preserving the intended constraints. The original failed attempt is retained in the audit record.

Scope: Linear elasticity and small-displacement equilibrium or linearized bifurcation/modal analysis; this is computational verification, not a material-test validation.

Reference and reproduction

Timoshenko and Gere, Theory of Elastic Stability, 2nd ed. (1961), plate buckling.

Wierzbicki, T. MIT 2.080J Structural Mechanics, Lecture 11: Buckling of Plates and Sections (2013). https://ocw.mit.edu/courses/2-080j-structural-mechanics-fall-2013/resources/mit2_080jf13_lecture11/

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv